1.a

We need to prove that for all P, Q, that the KL divergence between the two distributions is always nonnegative. Here, we use Jensen's inequality. We can easily see from the second derivative of $f(x) = -\log(x)$ ($f''(x) = \frac{1}{x^2}$ positive for all $x \in \mathbb{R}$) that $-\log(x)$ is a strictly convex function on its domain $x > 0$. We are even given from the problem hint that $f(x) = -log(x)$ is strictly convex. From this, we can establish the inequality that

$$-\log(E[X]) \leq E[-\log(x)]$$

We can see how $D_{KL}(P \mid\mid Q) = \sum_{x\in\mathcal{X}} P(x)\log\left(\frac{P(x)}{Q(x)}\right)$ is in the form $\underset{x\sim P}{E}\left[-\log\left(\frac{Q(x)}{P(x)}\right)\right]$. We can therefore say that

$$\begin{aligned}
D_{KL}(P \mid\mid Q) = \underset{x\sim P}{E}\left[-\log\left(\frac{Q(x)}{P(x)}\right)\right] &\geq -\log\left(\underset{x\sim P}{E}\left[\frac{Q(x)}{P(x)}\right]\right) = -\log\left(\sum_{x\in\mathcal{X}} P(X)\frac{Q(x)}{P(x)}\right) \\
&= -\log\left(\sum_{x\in\mathcal{X}} Q(x)\right) = -\log(1) = 0 \\
&\Rightarrow D_{KL}(P \mid\mid Q) \geq 0
\end{aligned}$$

Or in other words, the KL divergence of any two probability distribution P, Q is always nonnegative.

Now we need to prove that $D_{KL}(P \mid\mid Q) = 0 \Leftrightarrow P = Q$.

Assumption 1. $D_{KL}(P \mid\mid Q) = 0$
From the above solution, we get the following:

$$D_{KL}(P \mid\mid Q) = \underset{x\sim P}{E}\left[-\log\left(\frac{Q(x)}{P(x)}\right)\right] = 0$$

Now, $\log\left(\frac{Q(x)}{P(x)}\right) \in \mathbb{R}$ for $Q(x), P(x) > 0$. We know from that if we have a real number $n \in \mathbb{R}$,

$$\forall P. \quad \underset{x\sim P}{E}[n] = \sum_x P(x)\cdot n = n\sum_x P(x) = n\cdot 1 = n$$

Intuitively, no matter what the probability distribution for x is, the weighted average of a constant function is that constant. Therefore,

$$\underset{x\sim P}{E}\left[-\log\left(\frac{Q(x)}{P(x)}\right)\right] = 0 \Rightarrow -\log\left(\frac{Q(x)}{P(x)}\right) = 0 \Rightarrow \frac{Q(x)}{P(x)} = 1 \Rightarrow Q(x) = P(x)$$

We have proven that $D_{KL}(P \mid\mid Q) = 0 \Rightarrow P = Q$

Assumption 2. $P = Q$
Let's substitute Q for P in the KL divergence equation:

$$D_{KL}(P \mid\mid Q) = \sum_x P(x)\log\left(\frac{P(x)}{P(x)}\right) = \sum_x P(x)\log(1) = \sum_x P(x)\cdot 0 = 0$$

We have proven that $P = Q \Rightarrow D_{KL}(P \mid\mid Q) = 0$

Since we have proven the statement in both directions, we can therefore conclude that

$$D_{KL}(P \mid\mid Q) = 0 \quad \text{if and only if} \quad P = Q$$

1.b

We are given the following definition for the KL divergence between 2 conditional distributions $P(X \mid Y), Q(X \mid Y)$:

$$D_{KL}(P(X \mid Y) \mid\mid Q(X \mid Y)) = \sum_y P(y) \left(\sum_x P(x \mid y) \log \left(\frac{P(x \mid y)}{Q(x \mid y)} \right) \right)$$

We will now prove that the above is equivalent to $D_{KL}(P(X) \mid\mid Q(X)) + D_{KL}(P(Y \mid X) \mid\mid Q(Y \mid X))$

$$D_{KL}(P(Y \mid X) \mid\mid Q(Y \mid X)) = \sum_x P(x) \left(\sum_y P(y \mid x) \log \left(\frac{P(y \mid x)}{Q(y \mid x)} \right) \right)$$

$$D_{KL}(P(X) \mid\mid Q(X)) = \sum_x P(x) \log \left(\frac{P(x)}{Q(x)} \right)$$

$$\begin{aligned}
D_{KL}(P(X) \mid\mid Q(X)) + D_{KL}(P(Y \mid X) \mid\mid Q(Y \mid X)) &= \sum_x P(x) \log \left(\frac{P(x)}{Q(x)} \right) + \sum_x P(x) \left(\sum_y P(y \mid x) \log \left(\frac{P(y \mid x)}{Q(y \mid x)} \right) \right) \\
&= \sum_x P(x) \log \left(\frac{P(x)}{Q(x)} \right) + \sum_{x,y} P(x) P(y \mid x) \log \left(\frac{P(y \mid x)}{Q(y \mid x)} \right) \\
&= \sum_x P(x) \log \left(\frac{P(x)}{Q(x)} \right) \cdot \sum_y P(y \mid x) + \sum_{x,y} P(x) P(y \mid x) \log \left(\frac{P(y \mid x)}{Q(y \mid x)} \right)
\end{aligned}$$

Since all terms of $D_{KL}(P(X) \mid\mid Q(X))$ only depend on x, we can easily multiply it by the term $\sum_y P(y \mid x) = 1$

$$\begin{aligned}
&= \sum_{x,y} P(x) P(y \mid x) \cdot \left(\log \left(\frac{P(x)}{Q(x)} \right) + \log \left(\frac{P(y \mid x)}{Q(y \mid x)} \right) \right) \\
&= \sum_{x,y} P(x) P(y \mid x) \cdot \log \left(\frac{P(x) P(y \mid x)}{Q(x) Q(y \mid x)} \right) \\
&= \sum_{x,y} P(x, y) \cdot \log \left(\frac{P(x, y)}{Q(x, y)} \right) \\
&= D_{KL}(P(X, Y) \mid\mid Q(X, Y))
\end{aligned}$$

1.c

We need to prove that

$$\underset{\theta}{\text{argmin}} D_{KL}(\hat{P}(X) \mid\mid P(X;\theta)) = \underset{\theta}{\text{argmax}} \sum_{i=1}^{n} \log(P(x^{(i)};\theta))$$

We first expand the KL divergence term:

$$\begin{aligned} D_{KL}(\hat{P}(X) \mid\mid P(X;\theta)) &= \sum_x \hat{P}(x) \log\left(\frac{\hat{P}(x)}{P(x;\theta)}\right) \\ &= \sum_x \hat{P}(x) \left(\log(\hat{P}(x)) - \log(P(x;\theta))\right) \\ &= \sum_x \hat{P}(x) \log(\hat{P}(x)) - \sum_x \hat{P}(x) \log(P(x;\theta)) \end{aligned}$$

Since $\hat{P}(x) = \frac{1}{n}\sum_{i=1}^{n} \mathbf{1}\{x^{(i)} = x\}$, it doesn't change over the parameter θ. Now we can see that minimizing $D_{KL}(\hat{P}(X) \mid\mid P(X;\theta))$ over θ is equivalent to maximizing $\sum_x \hat{P}(x)\log(P(x;\theta))$. In other words,

$$\underset{\theta}{\text{argmin}} D_{KL}(\hat{P}(X) \mid\mid P(X;\theta)) = \underset{\theta}{\text{argmax}} \sum_x \hat{P}(x) \log(P(x;\theta))$$

Now we can see that $sum_{i=1}^{n} \log(P(x^{(i)};\theta))$ is the sum of the log-likelihoods and that $\sum_x \hat{P}(x)\log(P(x;\theta))$ is the expected log-likelihoods over the emperical distribution of x as defined by $\hat{P}(x)$. We can rewrite $\sum_x \hat{P}(x)\log(P(x;\theta))$ using the definition of $\hat{P}(x)$

$$\sum_x \hat{P}(x) \log(P(x;\theta)) = \sum_x \frac{1}{n} \sum_{i=1}^{n} \mathbf{1}\{x^{(i)} = x\} \log(P(x;\theta)) = \frac{1}{n} \sum_{i=1}^{n} \log(P(x^{(i)};\theta))$$

Intuitively, the left side is grouping the same values of x ($\mathbf{1}\{x^{(i)} = x\}$), counting how many times they appear ($\sum_{i=1}^{n}$), and multiplying the log-likelihood once by that count while the right side is going through each datapoint $x^{(i)}$ and finding the log-likelihood of each and summing them up. They are equivalent.

$$\underset{\theta}{\text{argmax}} \sum_x \hat{P}(x) \log(P(x;\theta)) = \underset{\theta}{\text{argmax}} \frac{1}{n} \sum_{i=1}^{n} \log(P(x^{(i)};\theta)) = \underset{\theta}{\text{argmax}} \sum_{i=1}^{n} \log(P(x^{(i)};\theta))$$

Therefore, we can conclude that

$$\underset{\theta}{\text{argmin}} D_{KL}(\hat{P}(X) \mid\mid P(X;\theta)) = \underset{\theta}{\text{argmax}} \sum_{i=1}^{n} \log(P(x^{(i)};\theta))$$

4.a

We are told to find the minimizer $\hat{\beta}_\lambda = \arg\min_{\beta \in \mathbb{R}^d} J_\lambda(\beta)$ for the regularized linear regression cost function

$$J_\lambda(\beta) = \frac{1}{2}||X\beta - \vec{y}||_2^2 + \frac{\lambda}{2}||\beta||_2^2$$

for when $\lambda > 0$. Expanding out, we get

$$J_\lambda(\beta) = \frac{1}{2}(X\beta - \vec{y})^T(X\beta - \vec{y}) + \frac{\lambda}{2}\beta^T\beta$$
$$\nabla_\beta J_\lambda(\beta) = X^T(X\beta - \hat{y}) + \lambda\beta$$

Setting the gradient equal to 0, we can find the minimum for the cost function:

$$X^T(X\beta - \hat{y}) + \lambda\beta = 0$$
$$X^T X\beta - X^T\hat{y} + \lambda\beta = 0$$
$$(X^T X + \lambda I_{d\times d})\beta = X^T\hat{y}$$
$$\Rightarrow \beta = (X^T X + \lambda I_{d\times d})^{-1}X^T\hat{y}$$

When lambda is positive, adding λI to the Gram matrix $X^T X$ shifts the eigenvalues so that the eigenvalues of $X^T X + \lambda$ becomes positive, ensuring the matrix $X^T X + \lambda I_{d\times d}$ becomes positive definite and invertible in order to solve for β. Therefore,

$$\hat{\beta}_\lambda = (X^T X + \lambda I_{d\times d})^{-1}X^T\vec{y}$$